

LLM Survey Framework: Coverage, Reasoning, Dynamics, Identification

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The views expressed herein are those of the authors and do not necessarily represent the views of the Federal Reserve Bank of Kansas City or the Federal Reserve System

Overview

- Brief summary:
 - Develop a method to simulate information treatment experiments in surveys using LLMs
 - Validate by replicating a series of RCTs on US consumer inflation expectations; confirm that treatment is less effective when inflation is high
 - Demonstrate LLM advantages: cost, sample period extension, follow-up questions on mechanisms, reasoning etc.
- High expected value project, but with variance
 - Potentially extremely valuable
 - With a major challenge: must clear a high bar to establish trustworthiness

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 - ... they forecast more accurately and *set prices/inputs more profitably*.
- But such studies are expensive. A cheap alternative would be great!

What are the challenges?

- Potential users (and referees) must be convinced that LLMs provide a reasonable substitute
- Do LLMs respond to information treatments in the same way as humans?
- Will methods that work in standard settings generalize?
- Are the timing restrictions effective?

Validation Idea 1: Formality

- A formal statistical model would help e.g.

consumer forecasts:

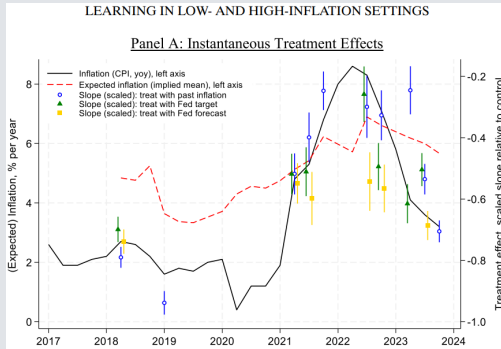
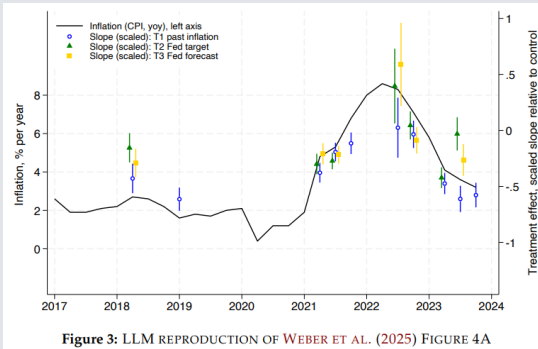
$$f_{it} = x_{it}\beta + u_{it}$$

LLMs model:

$$\hat{f}_{it} = x_{it}\hat{\beta}$$

- Should we expect $\hat{\beta}$ is unbiased?
- LLM output is stochastic; to what extent does it represent uncertainty about β versus u_{it} ?
- Heterogeneity is probably inaccurate but maybe average effects are okay?
- See Sanjog Misra's *Foundation Priors* (2025)
- Formal model is important for **inference**

Validation Idea 1: Formality



Are left and right panels estimating the same parameter?

A statistical framework + tests will help

Validation Idea 2: Overwhelming Variety

- Weber et al (2025) is a nice starting point. Do LLMs do well in dissimilar settings?
 - For people to apply to novel experiments, they will want to know that LLMs do well across many different types of existing experiments
 - You do this across time (good), but across environments would help
- Strategy of Colarieti, Mei, and Stantcheva (2024) is a good model
 - Colarieti et al want to know how people respond to income shocks (MPCs)
 - They experiment by giving survey respondents different hypothetical scenarios and ask for their consumption plans
 - Is this reasonable? They validate by simulating many different studies

Overwhelming Variety: Colarieti, Mei, and Stantcheva (2024)

TABLE 2: CROSS-VALIDATIONS

Paper	Estimate	Sample	Value	Our estimate
Patterson (2023)	MPC out of income loss due to unemp.	CEX, PSID	.53	.59 (.024)
Kaplan et al. (2014)	Share of HtM households	SCF	.31	.31 (.013)
	Share of wealthy HtM out of total HtM		.62	.64 (.036)
Chetty and Szeidl (2007)	Share of committed expenditures	CEX, PSID	0.5 (update: 0.6)	.62 (.005)
Baugh et al. (2021)	MPC out of tax refund, 30 days before receipt	Admin data, account aggregator	.001	.01 (.002)
	MPC out of tax refund, 30 days after receipt		.07	.091 (.009)
	MPC out of tax refund, 30-60 days after receipt		.03	.096 (.009)
Baugh et al. (2021)	MPC out of tax payment, 30 days before due	Admin data, account aggregator	.001	.044 (.007)
	MPC out of tax payment, 30 days after due		.001	.026 (.004)
	MPC out of tax payment, 30-60 days after due		.01	.02 (.004)
Di Maggio et al. (2017)	Car spending/initial mort. paym. out of cuts in mort. paym.	BlackBox Logic, Equifax	.043	.065 (.02)
	Repaym. of mortgage debt/initial mort. paym. out of cuts in mort. paym.		.043	.059 (.008)
Karger and Rajan (2021)	MPC out of the <u>first</u> EIP	Facteus bank-account data	.46	.5 (.024)
Misra et al. (2022)	MPC out of the <u>first</u> EIP	Facteus data, ZIP code level	.51	
Chetty et al. (2023)	MPC out of the <u>first</u> EIP	Affinity Solutions, aggregated data	.37-.61	

Notes. Cross-validations are taken from the first survey wave (May - October 2021), except Baugh et al. (2021), Di Maggio et al. (2017) that are based on the second cross-validation survey (February 2024). Karger and Rajan (2021) estimate the MPC out of the first EIP over an horizon of 14 days. They also show that spending responses are concentrated in the first week after the transfer receipt, while the response flattens already in the following week. Misra and Surico (2014) estimate the MPC over the first 4 days after the first EIP receipt. They also show that the spending response is flat after the first 6 days. Chetty et al. (2023) estimate the MPC over one-month after first EIP receipt separately for income quartiles. Their MPC ranges between 0.37 and 0.61 corresponding respectively to the MPCs in the lowest and highest quartile. Standard errors in parentheses.

Overwhelming Variety: Weber et al (2025)

TABLE I
OVERVIEW OF RCTs.

Country	Agents	RCT Dates	Priors	Posteriors	Information Treatments
United States	Households (~20 K per wave)	2018Q2, 2019Q1, 2021Q2–Q4, 2022Q3–Q4, 2023Q2–Q4	One-year-ahead inflation expectations from distribution	One-year-ahead inflation expectations from point forecast	• Inflation over the last year • FOMC inflation target • FOMC inflation forecast
Euro area	Households (~10 K per wave)	2021Q4, 2022Q2–Q2, 2022Q4	One-year-ahead inflation expectations from distribution	One-year-ahead inflation expectations from point forecast	• Inflation over the last year • ECB inflation target and past inflation • Professional inflation forecast • Inflation over the last year
Netherlands	Households (~2000)	2018Q2	One-year-ahead inflation expectations from distribution	One-year-ahead inflation expectations from point forecast	• Inflation over the last year
United States	Firms (~300 per wave)	2019Q1, 2023Q1	Perceived inflation over last year	One-year-ahead inflation expectations from point forecast	• Inflation over the last year
Uruguay	Firms (~500 per wave)	2018Q1–Q2, 2019Q2, 2023Q1	One-year-ahead inflation expectations from point forecast	One-year-ahead inflation expectations from next wave	• Inflation over the last year • Central Bank of Uruguay inflation target range
New Zealand	Firms (~2000 per wave)	2014Q4, 2016Q2, 2018Q1, 2019Q3	One-year-ahead inflation expectations from distribution	One-year-ahead inflation expectations from point forecast	• Inflation over the last year • Reserve Bank of NZ inflation target • Professional forecast of inflation • Combination
Italy	Firms (~1000 per wave)	2012Q3– 22Q4	Inflation expectations in previous quarter from point forecast	One-year-ahead inflation expectations from point forecast	• Inflation over the last year in Italy and euro area

Overwhelming Variety: Other Topics

- Inflation is an obvious starting point, but info treatment RCTs exist for other expectations
- House prices: Chopra, Roth, and Wohlfart (2025)
- Exchange rates: Delgado, Herrero, Hofstetter, and Pedemonte (2024)
- Asset prices: Beutel and Weber (2025)
- See Coibion/Gorodnichenko book for many more examples

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- What are analogous questions for LLMs?

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 - Would be cool to see even more extrapolative applications... We got 1990s USA, but what about Weimar Germany?
- Convincing validation critical for use, take-up, publication
 - Might be a lot of work. Use AI!